



CV Project Requirements

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Project Requirements:

- Develop a computer vision (CV) application that processes visual data, such as **images, videos, or live camera streams**.
- The solution should effectively addresses a **real-world problem** using computer vision techniques(Yolo, CV2, VLLM,...)
- Deploy the trained model as a **cloud-hosted API**. The application must consume the model through API requests rather than including the model weights within the project.(Optional)
- Create a **user-friendly interface** using a framework such as **Streamlit** or **Gradio**.
- Ensure the application is **fully testable**, with clear instructions and sample inputs for evaluation.
- Record a short demonstration video showcasing the application's features and include it in the project's **GitHub README**.

Project Ideas

1. Plant Species Detection 🌿

Problem Statement: A plant nursery or garden center wants to develop an application that helps new employees identify different plant species accurately. Since many staff members have limited botanical knowledge, the system should automate the identification process and reduce human errors.

The application should allow users to:

- Upload an image of a plant or capture one using their device's camera.
- Detect the plant in the image using a YOLO object detection model.
- Draw a bounding box around each detected plant.
- Display the predicted plant species and the model's confidence score.
- Optionally, provide basic information or care instructions for the detected species.

Dataset: <https://universe.roboflow.com/houseplants-dataset-for-bibitunggulid-bangkit-2023/houseplants-image-detection>

Project Ideas

2. Smart Security Alarm System 🚨

Problem Statement: A jewelry store wants to monitor its premises outside business hours. The goal is to automatically detect unauthorized human presence and immediately notify the owner to reduce the risk of theft or intrusion.

The application should:

- Monitor a live camera feed or uploaded surveillance video.
- Detect people using a YOLO object detection model.
- Trigger an alarm whenever a person is detected during non-working hours.
- Send a notification (email, SMS, or Telegram) to the store owner.
- Save snapshots of detected intrusions with timestamps for future review.

Dataset: <https://universe.roboflow.com/cctvindoorperson/cctv-indoor-person-detection>

Project Ideas

3. Personal Protective Equipment (PPE) Safety Monitoring 🏗️

Problem Statement: Construction companies and industrial facilities need to ensure that workers wear the required Personal Protective Equipment (PPE) before entering hazardous work areas. Manual monitoring is time-consuming and prone to human error.

The application should:

- Process images, videos, or live camera streams.
- Detect workers and identify whether they are wearing the required PPE (e.g., helmets, safety vests, gloves, and masks).
- Highlight workers who are missing mandatory safety equipment.
- Display real-time safety statistics, such as(optional):
 - Total workers detected
 - Safe workers
 - Workers missing helmets
 - Workers missing safety vests

Dataset: <https://universe.roboflow.com/sdp-lfigk/ppe-detection-ozhfb/browse?queryText=&pageSize=50&startingIndex=0&browseQuery=true>

GitHub Repository Requirements:

- Include a clear and comprehensive description of the project.
- Provide a professional README that explains the project's purpose, installation, usage, and execution steps.
- Include appropriate technical documentation for the project.
- Follow Git best practices for version control, including meaningful commit messages and proper branch management.
- Mention the training program under which the project was developed.
- Include a reference to the SDAIA Academy GitHub account when applicable: <https://github.com/SDAIAAcademy>